Deep Learning for Extreme Weather Forecasting: A Survey

Extreme weather forecasting is a high-impact application of deep learning because severe precipitation, tropical cyclones, heat waves, floods, atmospheric rivers, droughts, wildfires, and damaging wind events are rare, high-loss phenomena that stress both numerical weather prediction and statistical post-processing. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around global data-driven weather models, precipitation nowcasting, extreme-event detection, spatiotemporal architectures, data assimilation, downscaling, probabilistic forecasting, remote sensing datasets, evaluation, robustness, physical consistency, and operational deployment. The survey argues that progress requires not only lower average forecast error but also calibrated tail risk, physically plausible trajectories, event-centered verification, and trustworthy integration with meteorological workflows.

Keywords:

deep learning; extreme weather; weather forecasting; precipitation nowcasting; numerical weather prediction; uncertainty; remote sensing; climate risk

1 Introduction

Extreme weather forecasting sits at the intersection of atmospheric science, machine learning, decision support, and risk management. The events of greatest societal concern - heavy precipitation, flash floods, tropical cyclones, heatwaves, atmospheric rivers, droughts, severe convective storms, wildfire weather, hail, and damaging winds - are rare, spatially heterogeneous, and strongly coupled to local exposure. A model that improves average temperature or geopotential-height error may still fail the forecast that matters most: the tail event with short lead time, sharp spatial gradients, and costly consequences [37, 101, 153, 178, 191, 270, 280, 299].

Deep learning has changed the technical landscape of weather prediction. Neural systems now learn global atmospheric dynamics from reanalysis, emulate portions of numerical weather prediction, assimilate multimodal data, produce precipitation nowcasts from radar and satellite imagery, downscale coarse fields, calibrate ensembles, and estimate uncertainty. The strongest systems combine large-scale spatiotemporal representation learning with meteorological structure: variables live on grids or meshes, dynamics are advective and multiscale, and forecasts must remain stable over long rollouts [101, 153, 178, 191, 221, 270, 280, 299].

Extreme-weather forecasting requires additional care. Extreme labels are sparse, observations are noisy, and common loss functions overemphasize ordinary conditions. Evaluation must therefore ask whether a model detects the event, localizes its footprint, estimates intensity, gives calibrated uncertainty, preserves physical consistency, and supports decisions under lead-time constraints. This survey reviews the deep-learning literature from that event-centered perspective.

2 Background

2.1 Weather Forecasting and Extreme Events

Operational meteorology traditionally relies on numerical weather prediction, data assimilation, ensemble simulation, statistical post-processing, expert interpretation, and verification. Extreme events occupy the tails of these systems. They may be driven by mesoscale convection, land-atmosphere feedback, tropical cyclone dynamics, blocking patterns, atmospheric rivers, compound hazards, or local topography. Their rarity creates a statistical tension: models need vast training corpora, yet the most important outcomes are underrepresented in the loss landscape [101, 145, 153, 178, 191, 270, 280, 299].

2.2 Deep Learning in the Forecasting Pipeline

Deep learning can enter the pipeline at several points. End-to-end data-driven models map recent atmospheric states to future states. Nowcasting models extrapolate radar or satellite sequences for minutes to hours. Hybrid systems correct or downscale numerical forecasts. Data-assimilation models learn observation-to-state mappings. Probabilistic models generate ensembles or predictive distributions. Event detectors convert fields into warnings or risk maps. These roles are complementary rather than mutually exclusive.

2.3 Why Extremes Are Different

Extreme weather is not simply the upper quantile of a scalar target. The event definition may depend on location, season, duration, antecedent conditions, and impact. A flood forecast depends on precipitation, soil moisture, terrain, hydrology, and exposure. A heatwave depends on temperature, humidity, nighttime minima, persistence, and population vulnerability. A tropical cyclone forecast involves track, intensity, size, rainfall, surge, and uncertainty. Deep models for extremes should therefore align prediction targets with event mechanisms and decision thresholds.

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined deep learning, machine learning, neural networks, transformers, graph neural networks, neural operators, diffusion models, weather forecasting, numerical weather prediction, global and medium-range forecasts, precipitation nowcasting, radar, satellite, remote sensing, ERA5, WeatherBench, SEVIR, data assimilation, downscaling, probabilistic forecasting, verification, and event terms such as extreme weather, severe storms, tropical cyclones, hurricanes, heatwaves, floods, droughts, atmospheric rivers, wildfire weather, hail, and storm surge. Records were retained when title or abstract evidence explicitly linked meteorological forecasting with deep-learning or machine-learning methods and with either extreme events, nowcasting, forecast datasets, uncertainty, post-processing, or operationally relevant weather prediction. The final bibliography cites 300 unique scholarly arXiv preprints.

3.2 Taxonomy

We organize the literature along eight axes. The first is *forecast horizon*: minutes-to-hours nowcasting, short-range forecasts, medium-range global forecasts, subseasonal prediction, and climate-scale risk projection. The second is *hazard*: precipitation, convection, tropical cyclone, heat, flood, drought, atmospheric river, wildfire weather, hail, wind, and compound events. The third is *data source*: reanalysis, numerical model output, radar, satellite, station observations, lightning, hydrological data, or impact labels. The fourth is *architecture*: convolutional recurrent networks, U-Nets, transformers, graph neural networks, neural operators, diffusion models, and foundation models. The fifth is *learning objective*: deterministic regression, classification, quantile prediction, distribution learning, ensemble generation, anomaly detection, or decision loss. The sixth is *physics interface*: coordinate systems, conservation constraints, differentiable dynamics, assimilation, hybrid correction, and post-processing. The seventh is *evaluation*: pointwise error, event skill, spatial overlap, calibration, reliability, lead-time performance, and tail metrics. The eighth is *deployment*: latency, interpretability, robustness, domain shift, forecaster trust, and warning workflow integration.

3.3 Global Data-Driven Weather and Climate Foundation Models

Global data-driven systems learn atmospheric evolution directly from reanalysis and model fields. They can produce fast medium-range forecasts and may serve as backbones for downstream extreme-event prediction. Their central strength is scale: they learn multivariate, multi-level, global dynamics from long historical archives. Their central risk is averaging: a model optimized for aggregate skill may smooth extremes, underrepresent rare regimes, or drift physically during long autoregressive rollouts [101, 153, 178, 191, 221, 270, 280, 299].

3.4 Nowcasting and Short-Range Precipitation Forecasting

Nowcasting is crucial for flash floods, convective storms, aviation hazards, and urban decision support. Radar and geostationary satellite sequences provide high-resolution observations, but precipitation evolves through advection, growth, decay, splitting, and merging. Deep nowcasting models learn these patterns using convolutional recurrent networks, attention, generative models, and optical-flow-inspired structures. Evaluation must penalize both blurry averages and overconfident false alarms [1, 18, 101, 175, 178, 191, 270, 293].

3.5 Extreme-Event Detection and Impact-Oriented Forecasting

Event-centered forecasting converts atmospheric fields into hazard information. For tropical cyclones, models may estimate genesis, track, intensity, wind radii, or rainfall. For atmospheric rivers, they may segment filaments and predict landfall. For heatwaves and droughts, persistence and land-surface memory matter. For floods and wildfire weather, meteorological forecasts must connect to hydrological or fuel conditions. Deep learning can help by extracting spatial patterns, fusing variables, and learning impact proxies, but event definitions and labels must be explicit [101, 145, 153, 178, 191, 270, 280, 299].

3.6 Spatiotemporal Architectures and Neural Operators

Weather fields are high-dimensional, multiscale, and geographically structured. Convolutional networks capture local patterns; recurrent networks model temporal memory; transformers capture long-range dependencies; graph neural networks handle spherical meshes and irregular observations; neural operators learn mappings between function spaces; diffusion and generative models can represent forecast distributions. Architecture choice should reflect both atmospheric dynamics and operational constraints such as rollout stability, resolution, and latency [37, 145, 160, 178, 191, 221, 270, 299].

3.7 Data Assimilation, Downscaling, and Post-Processing

Deep learning is often most useful when coupled with existing forecast systems. Learned data assimilation can map sparse or noisy observations into latent atmospheric states. Downscaling can convert coarse forecasts to local fields. Super-resolution can sharpen spatial structure. Bias correction and post-processing can calibrate numerical or neural forecasts. These hybrid roles are especially important for extremes because local terrain, sensor coverage, and systematic model bias can dominate event-level skill [1, 18, 37, 92, 153, 164, 178, 280].

3.8 Probabilistic Forecasting, Uncertainty, and Evaluation

Extreme-weather decisions require probabilities, not only point estimates. A warning system must communicate the chance of exceeding a threshold, the plausible event footprint, and the uncertainty over timing and intensity. Deep models use ensembles, latent-variable models, quantile losses, diffusion models, Bayesian approximations, and calibration layers to represent uncertainty. Verification should include reliability diagrams, Brier scores, CRPS, rank histograms, precision, recall, spatial neighborhood scores, and rare-event metrics [18, 37, 101, 164, 191, 221, 270, 299].

3.9 Remote Sensing, Multimodal Data, and Benchmarks

Remote-sensing data are central for short-lead extremes. Radar resolves precipitation structure, satellites observe clouds and moisture over broad domains, lightning and station networks provide complementary signals, and reanalysis archives provide global context. Benchmark datasets such as WeatherBench-style reanalysis tasks and SEVIR-style severe weather imagery have helped standardize comparisons. However, benchmark design must avoid rewarding only easy regimes; tail events, missing sensors, regional differences, and temporal leakage need explicit treatment [37, 101, 145, 153, 221, 270, 280, 299].

3.10 Robustness, Physical Consistency, and Operational Deployment

Operational deployment requires stability under distribution shift. Climate change, sensor upgrades, changing observing systems, regional transfer, rare compound events, and novel extremes can all move the test distribution. Physical consistency matters because forecasts should preserve plausible mass, moisture, energy, and motion patterns. Deployment also requires interpretability, latency constraints, fallback behavior, monitoring, and integration with human forecasters rather than isolated benchmark scores [37, 101, 145, 153, 160, 164, 270, 280].

3.11 Representative Literature Map

The following map cites the retained corpus by theme, making the survey coverage explicit across global weather models, nowcasting, extreme-event detection, spatiotemporal architectures, assimilation and downscaling, probabilistic forecasting, remote sensing, benchmarks, robustness, and deployment.

Global data-driven weather and climate foundation models The global data-driven weather and climate foundation models cluster contains work that develops methods, systems, datasets, or analyses central to this survey [11, 18, 101, 145, 164, 191, 221, 280, 299]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [1, 27, 57, 68, 112, 151, 175, 209, 293]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [92, 100, 114, 126, 158, 201, 202, 243, 246]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [9, 58, 61, 88, 143, 177, 190, 266, 295]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [43, 73, 102, 182, 189, 206, 224, 264, 291]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [41, 53, 64, 103, 140, 186, 195, 214, 220]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [76, 79, 154, 218, 234, 240, 286, 288, 298]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [31, 81, 111, 131, 156, 181, 236, 263, 297]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [38, 44, 56, 174, 253, 259, 272, 279, 294]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [19, 26, 29, 34, 36, 42, 161, 179, 184]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [52, 55, 69, 121, 138, 166, 249, 252, 292]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [45, 51, 54, 62, 128, 196, 199, 203, 277]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [5, 46, 47, 89, 104, 115, 169, 225, 269]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [125, 132, 135, 137, 146, 149, 172, 244, 268]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 16, 67, 120, 157, 231, 232, 254, 282]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [23, 49, 122, 133, 251, 265, 278, 289, 296]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [10, 15, 17, 185, 208, 242, 261, 276, 285]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [48, 82, 212, 215, 238, 274].

Nowcasting and short-range precipitation forecasting The nowcasting and short-range precipitation forecasting cluster contains work that develops methods, systems, datasets, or analyses central to this survey [7, 65, 105, 148, 192, 228, 229, 237, 275]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [14, 72, 77, 96, 99, 107, 188, 219, 248]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [6, 35, 40, 50, 116, 136, 176, 210, 284]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [21].

Extreme-event detection and impact-oriented forecasting The extreme-event detection and impact-oriented forecasting cluster contains work that develops methods, systems, datasets, or analyses central to this survey [2, 13, 20, 32, 39, 59, 86, 98, 155]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [90, 117, 119, 150, 163, 171, 256, 257, 300]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [63, 66, 80, 93, 127, 141, 162, 204, 222]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [205].

Spatiotemporal architectures and neural operators The spatiotemporal architectures and neural operators cluster contains work that develops methods, systems, datasets, or analyses central to this survey [97, 142, 147, 167, 178, 194, 233, 239, 258]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [4, 8, 84, 94, 123, 207, 245, 255, 262]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [85].

Data assimilation, downscaling, and post-processing The data assimilation, downscaling, and post-processing cluster contains work that develops methods, systems, datasets, or analyses central to this survey [37, 113, 129, 153, 159, 173, 230, 271, 290]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [28, 33, 71, 83, 95, 118, 134, 267, 287]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [106, 139].

Probabilistic forecasting, uncertainty, and evaluation The probabilistic forecasting, uncertainty, and evaluation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [25, 30, 78, 91, 200, 211, 217, 270, 273]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [22, 75, 110, 144, 165, 168, 193, 216, 260]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [60, 187, 247].

Remote sensing, multimodal geoscience data, and datasets The remote sensing, multimodal geoscience data, and datasets cluster contains work that develops methods, systems, datasets, or analyses central to this survey [24, 70, 74, 124, 130, 160, 183, 250, 283]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [108, 170, 198, 213, 223, 226, 227, 235, 241]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [12, 87, 109, 152, 180, 281].

Robustness, physical consistency, and operational deployment The robustness, physical consistency, and operational deployment cluster contains work that develops methods, systems, datasets, or analyses central to this survey [197].

4 Challenges and Prospects

4.1 Tail-Aware Objectives

Mean-squared error and aggregate anomaly-correlation scores can hide failures on rare but important events. Future models should use objectives and sampling strategies that emphasize extremes, thresholds, spatial objects, and decision-relevant losses while preserving ordinary-weather skill.

4.2 Event Definitions and Labels

Extreme events are defined by hazards, locations, durations, impacts, and thresholds. Community datasets should record how events are labeled and should support multiple definitions, such as meteorological intensity, return period, warning category, and observed impact.

4.3 Calibration Under Rare Events

Forecast probabilities are useful only if they are calibrated in the tail. Deep systems need uncertainty methods that remain reliable for rare regimes and long lead times, including regional reliability analysis and evaluation under distribution shift.

4.4 Hybrid Physical-Neural Forecasting

The most useful systems may combine neural speed and representation learning with numerical models, data assimilation, physical constraints, and expert workflow. Hybrid systems can use neural models for emulation, correction, downscaling, ensemble generation, and event detection rather than replacing every part of the pipeline.

4.5 Operational Trust and Human Workflow

Extreme-weather warnings involve consequences for emergency managers, energy systems, transport, agriculture, and public health. Models should provide interpretable evidence, uncertainty, failure alerts, provenance, and comparison to baseline forecasts so that human forecasters can judge when to trust or override them.

5 Conclusion

Deep learning has become a major force in weather forecasting, from global data-driven models to radar nowcasting, forecast post-processing, downscaling, probabilistic prediction, and extreme-event detection. For extreme weather, the core challenge is not only forecasting the atmosphere but forecasting rare, high-impact events in a way that is calibrated, physically plausible, robust, and useful for decisions. Progress will depend on event-centered benchmarks, tail-aware objectives, multimodal observations, hybrid physical-neural workflows, and verification protocols that reward operational value rather than only average-field accuracy.

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